

Futureproof

EPM v14a

Constrained Parameter Optimisation

v2 — Corrected Constraints

March 2025

CONFIDENTIAL — For Internal Use Only

Constraints & Parameters

This section documents every constraint and parameter used in the optimisation. All scenarios are subject to these bounds — no configuration that violates a constraint is included in the results.

Hard Constraints

#	Constraint	Bound	Rationale
C 1	Collar width	$\leq \pm 20\%$	Wider collars too costly to hedge in practice
C 2	FP margin	$\geq 0.25\%$ (25 bps) p.a.	Commercial floor — minimum FP return
C 3	IO annuity $\geq$ PI annuity	Product design rule	Analysis uses identical ranges for like-for-like comparison
C 4	Total annuity is FIXED	Does not increase with longer term	Borrower chooses how to spread a fixed total over time
C 5	PoC — Reinsurance	$\leq 5\%$	Portfolio-level reinsurance pricing threshold
C 6	PoC — Individual LMI	$\leq 12\%$	Individual mortgage insurance pricing threshold

Fixed Model Parameters (not optimised)

Parameter	Value	Notes
Property value	\$2,000,000	Reference property
LVR	80%	
Gross mortgage	\$1,600,000	80% LVR $\times$ \$2M
Initial mortgage	Dynamic	\$1,600,000 – total annuity (e.g. \$1,350,000 at \$250K annuity)
Mortgage term	30 years	
Wholesale margin	2.00% p.a.	Funder cost of funds spread
Hedging fee	0.25% p.a.	Collar hedging cost on investment account
Retail margin	0.70% p.a.	Lender retail margin on mortgage balance
LMI upfront	1.60% of peak mortgage	Lenders mortgage insurance
Reinsurance upfront	0.10% of peak mortgage	Portfolio reinsurance layer
Equity return (mean)	10.0% p.a.	~70% S&P; 500, ~30% fixed income
Equity volatility	10.0% p.a.	Blended portfolio vol
Cash rate (initial)	4.40%	
Cash rate (long-run)	4.40%	Ornstein-Uhlenbeck mean reversion
Cash rate (kappa)	0.80	Mean reversion speed
Cash rate (sigma)	1.50%	Cash rate vol

Parameter	Value	Notes
Equity-rate correlation	0.069	
Holiday exit/entry ratio	1.62×	Fixed ratio

Optimised Parameters (search grid)

Parameter	Values Tested	Notes
Mortgage type	P&I, IO	P&I; amortises to \$0 at maturity; IO does not
Total annuity (both)	\$250,000 / \$275,000 / \$300,000 / \$350,000 / \$400,000	1.25%–2.00% of property value — identical range for like-for-like comparison
Payout term	10, 15, 20, 25 years	Annual draw = total ÷ term (total stays fixed)
Collar width	±10%, ±15%, ±20%	Symmetric cap/floor on equity returns
Holiday entry threshold	0.90, 0.95, 1.00, 1.05	As multiple of initial mortgage balance
Profit share %	20%, 25%	Share of surplus extracted for FP
Profit share period	Every 3yr, Every 5yr	5yr preferred (better compounding)
FP margin	0.25%, 0.30%, 0.35%	Applied to investment account balance p.a.

Annuity Structure (Critical)

The **total annuity** is set by the annuity rate and is **fixed**. Longer payout terms spread the same total over more years, reducing the annual draw. However, the **peak mortgage is the same** for all terms (\$1.6M) because initial loan + total annuity = gross loan regardless of how the annuity is spread. For P&I, shorter payout terms leave more years for amortisation after the annuity period ends, which generally produces **better PoC**.

Total Annuity	Rate	10yr	15yr	20yr	25yr
\$250,000	1.25%	\$25,000/yr	\$16,667/yr	\$12,500/yr	\$10,000/yr
\$275,000	1.375%	\$27,500/yr	\$18,333/yr	\$13,750/yr	\$11,000/yr
\$300,000	1.50%	\$30,000/yr	\$20,000/yr	\$15,000/yr	\$12,000/yr
\$350,000	1.75%	\$35,000/yr	\$23,333/yr	\$17,500/yr	\$14,000/yr
\$400,000	2.00%	\$40,000/yr	\$26,667/yr	\$20,000/yr	\$16,000/yr

Simulation Methodology

Phase	Scenarios	Monte Carlo Paths	Purpose
Phase 1 — Individual levers	76	10,000	Isolate impact of each parameter
Phase 2 — Combined grid	5760	10,000	Full factorial search within constraints
Phase 3 — Validation	32	50,000	High-precision validation of top candidates
Total	5,868		

Random seed: 42. All phases use identical seed for reproducibility.

Executive Summary

With corrected constraints (collar $\leq \pm 20\%$, FP margin $\geq 0.25\%$, fixed total annuity), the optimisation tested 5,836 scenarios and validated 32 at 50,000 paths. Of the validated configurations, **5 met the reinsurance threshold** (PoC $\leq 5\%$) and **31 met the LMI threshold** (PoC $\leq 12\%$).

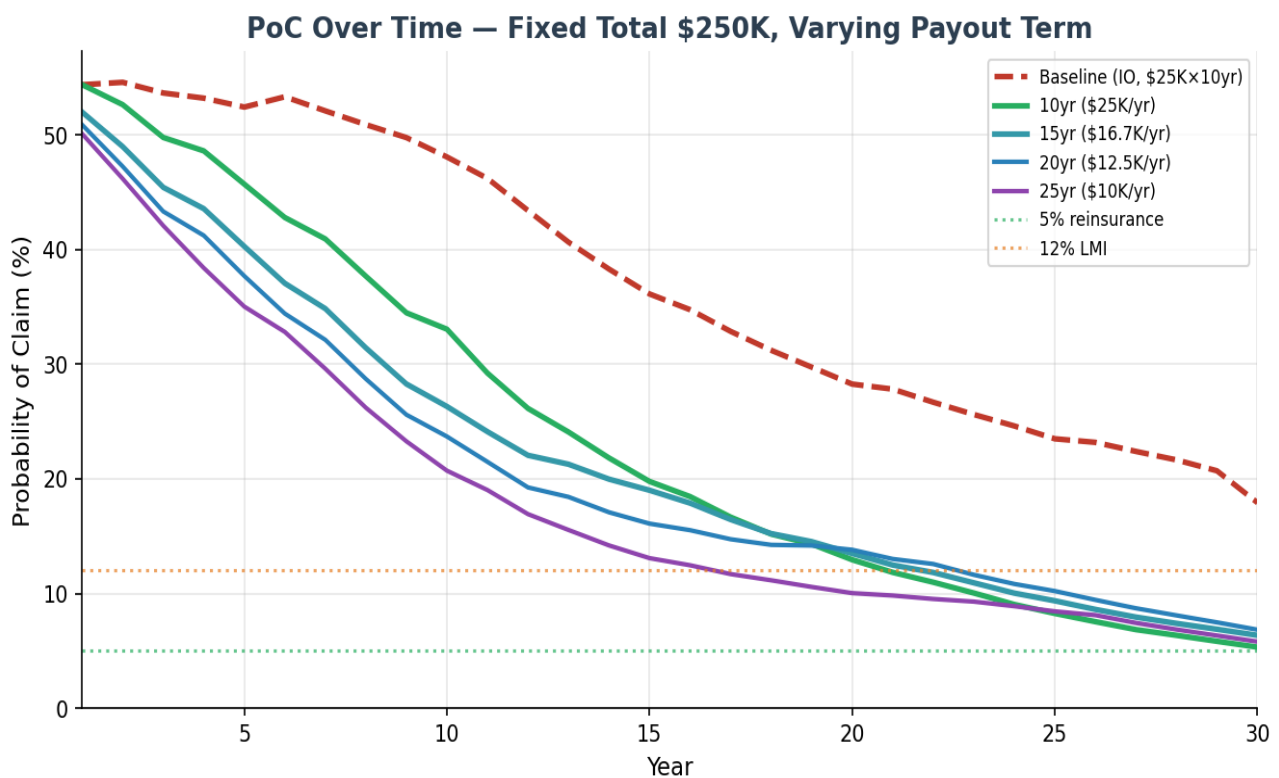
Metric	v14a Baseline	Reinsurance ($\leq 5\%$)	LMI ($\leq 12\%$)
Total Annuity	\$250,000	\$250,000	\$250,000
Payout	10yr @ \$25,000	10yr @ \$25,000	10yr @ \$25,000
Mortgage Type	IO	PI	PI
Collar	$\pm 20\%$	$\pm 20\%$	$\pm 20\%$
Holiday Entry	0.90	1.05	1.05
PS Config	25% q5	20% q3	25% q3
FP Margin	0.25%	0.25%	0.25%
PoC Year 30	17.9%	4.9%	5.3%
FP Revenue	\$823,670	\$1,428,338	\$1,505,910
Mean Surplus	\$1,381,981	\$1,440,007	\$1,078,273
Sharpe	0.829	1.333	1.363
Premium	\$44,472	\$12,485	\$12,956
Protection	82.1%	95.1%	94.7%
Final Mortgage	\$1,600,000	\$0	\$0

The key structural change from baseline is **P&I amortisation** (mortgage repays to \$0) and **holiday entry at 1.05** (more aggressive capital preservation). The collar remains at $\pm 20\%$ (same as baseline) and FP margin remains at 0.25% (same as baseline). The main revenue improvement comes from switching profit share to **every 3 years** instead of every 5.

Payout Duration Analysis

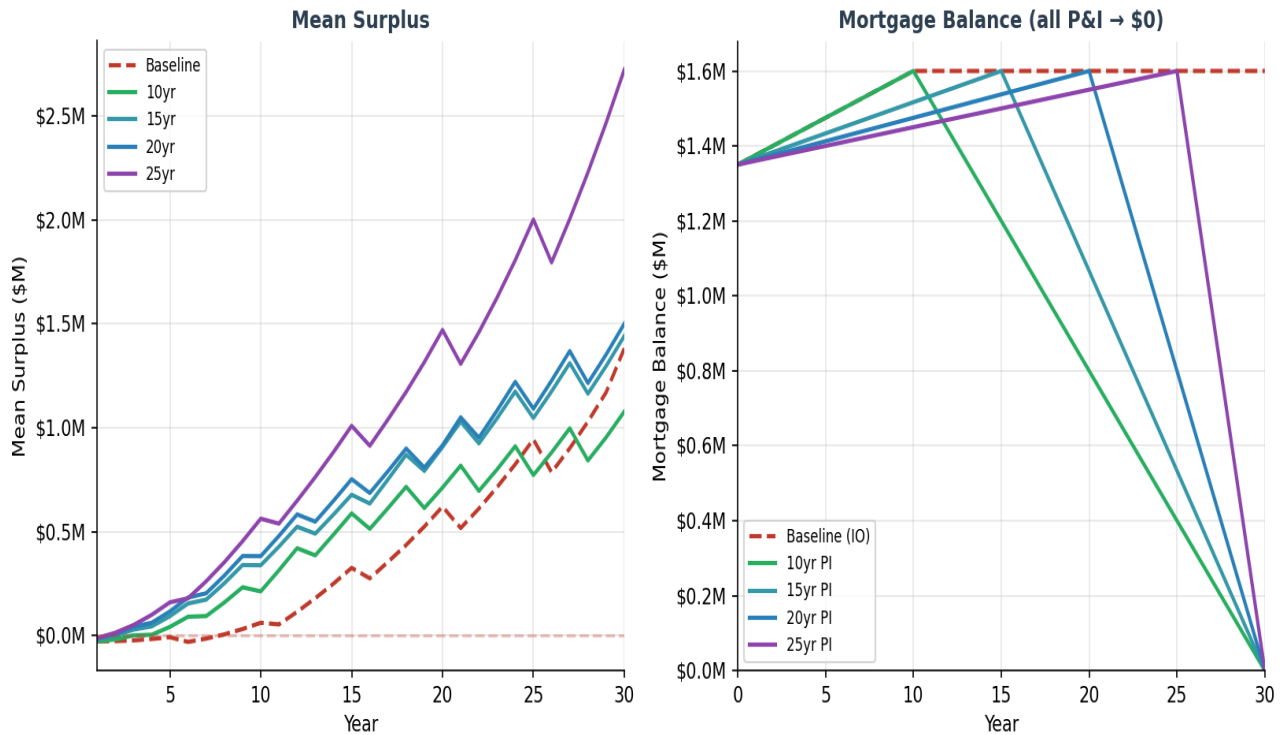
Since total annuity is fixed, all payout terms reach the same peak mortgage (\$1.6M). The key difference is **when amortisation begins**: a 10-year payout finishes drawing at year 10, leaving 20 years of P&I repayment. A 25-year payout doesn't start amortising until year 25, leaving only 5 years. Shorter payout terms therefore tend to produce better PoC for P&I mortgages.

Term	Total	Annual Draw	PoC	Tier	FP Revenue	Sharpe
10yr	\$250,000	\$25,000/yr	5.3%	LMI ✓	\$1,505,910	1.363
15yr	\$250,000	\$16,666/yr	6.3%	LMI ✓	\$1,499,738	1.202
20yr	\$250,000	\$12,500/yr	6.8%	LMI ✓	\$1,572,006	1.163
25yr	\$250,000	\$10,000/yr	5.8%	LMI ✓	\$1,287,243	1.180



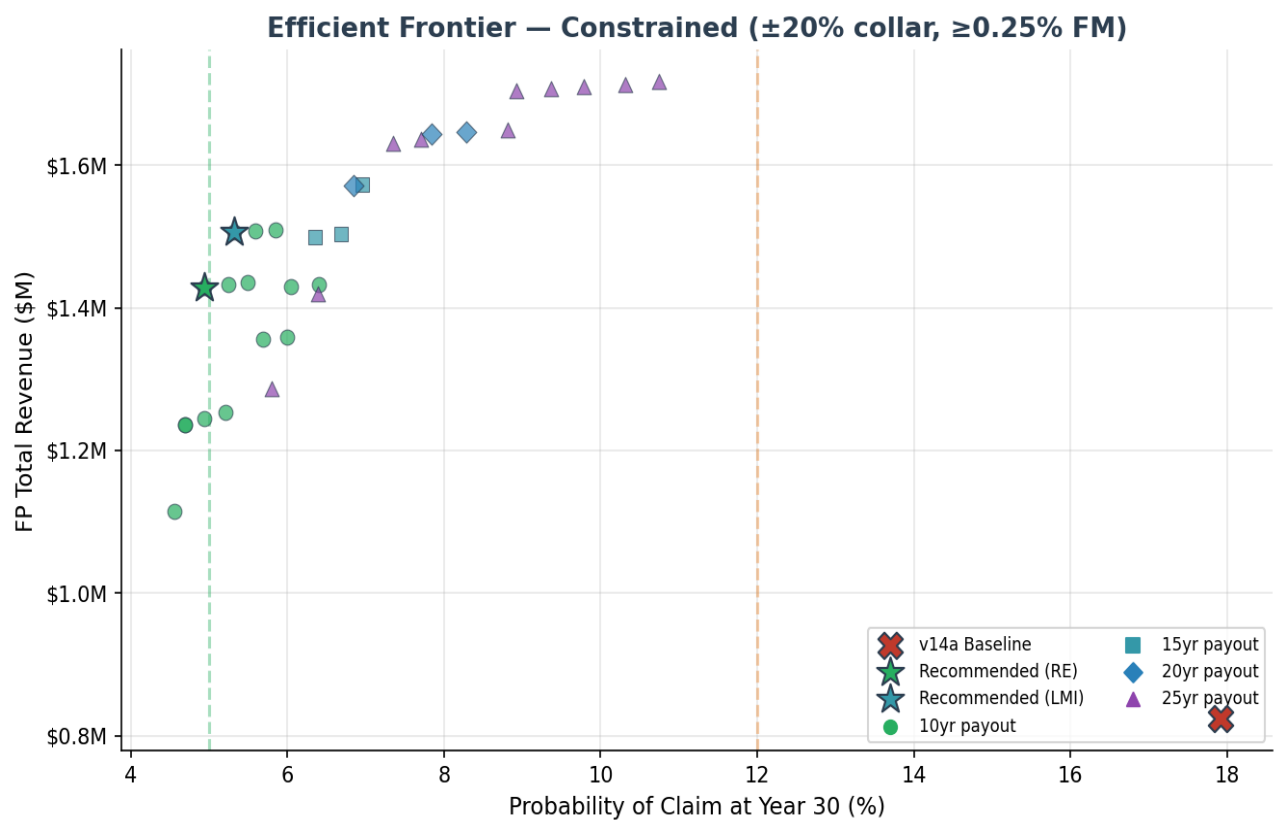
All optimised configurations dramatically outperform the baseline. The 10-year payout (green) achieves the **lowest** terminal PoC because the annuity period ends at year 10, leaving 20 years of P&I amortisation to reduce the mortgage. Longer payouts delay the amortisation phase, resulting in higher PoC despite the lower annual draw.

Surplus & Mortgage Trajectories



Left: Longer payout terms build higher surplus because the lower annual draw leaves more capital in the investment account to compound. Right: All P&I configurations amortise to \$0. All terms peak at the same \$1.6M, but the 10-year payout peaks at year 10 then amortises over 20 years; the 25-year payout peaks at year 25 then amortises over just 5 years.

Efficient Frontier



Tier 1 — Reinsurance Grade (PoC ≤ 5%)

5 configuration(s) meet the reinsurance threshold at 50K paths:

#	Configuration	Type	Payout	PoC %	FP Revenue	Sharpe
1	PI_12.50%_T\$250K_10y_C±19_HE=1.05_PS=20q ★	PI	\$250K/10yr	4.9%	\$1,428,338	1.333
2	PI_12.50%_T\$250K_10y_C±19_HE=1.05_PS=25q	PI	\$250K/10yr	4.7%	\$1,236,176	1.294
3	PI_\$250K_10y_HE=1.05	PI	\$250K/10yr	4.7%	\$1,236,176	1.294
4	PI_12.50%_T\$250K_10y_C±19_HE=1.05_PS=25q	PI	\$250K/10yr	4.9%	\$1,244,570	1.283
5	PI_12.50%_T\$250K_10y_C±19_HE=1.05_PS=20q	PI	\$250K/10yr	4.5%	\$1,114,228	1.254

★ = Recommended

Tier 2 — LMI Grade (PoC ≤ 12%)

31 configurations meet the LMI threshold. Top 25:

#	Configuration	Type	Payout	Annual	PoC %	FP Revenue	Sharpe
1	PI_12.50%_T\$250K_10y_C±19_HE=1.05_PS=2 ★	PI	\$250K/10yr	\$25,000/yr	5.3%	\$1,505,910	1.363
2	PI_12.50%_T\$250K_10y_C±19_HE=1.05_PS=2 ★RE	PI	\$250K/10yr	\$25,000/yr	4.9%	\$1,428,338	1.333
3	PI_12.50%_T\$250K_10y_C±19_HE=1.05_PS=2	PI	\$250K/10yr	\$25,000/yr	5.6%	\$1,507,434	1.344
4	PI_12.50%_T\$250K_10y_C±19_HE=1.05_PS=2	PI	\$250K/10yr	\$25,000/yr	5.2%	\$1,431,877	1.319
5	PI_12.50%_T\$250K_10y_C±19_HE=1.05_PS=2	PI	\$250K/10yr	\$25,000/yr	5.8%	\$1,509,251	1.323
6	PI_12.50%_T\$250K_10y_C±19_HE=1.05_PS=2	PI	\$250K/10yr	\$25,000/yr	5.5%	\$1,435,504	1.304
7	PI_12.50%_T\$250K_10y_C±19_HE=1.05_PS=2	PI	\$250K/10yr	\$25,000/yr	4.7%	\$1,236,176	1.294
8	PI_\$250K_10y_HE=1.05	PI	\$250K/10yr	\$25,000/yr	4.7%	\$1,236,176	1.294
9	PI_12.50%_T\$250K_10y_C±19_HE=1.05_PS=2	PI	\$250K/10yr	\$25,000/yr	4.9%	\$1,244,570	1.283
10	PI_13.75%_T\$275K_10y_C±19_HE=1.05_PS=2	PI	\$275K/10yr	\$27,500/yr	6.0%	\$1,430,204	1.305
11	PI_12.50%_T\$250K_10y_C±19_HE=1.05_PS=2	PI	\$250K/10yr	\$25,000/yr	4.5%	\$1,114,228	1.254

#	Configuration	Type	Payout	Annual	PoC %	FP Revenue	Sharpe
12	PI_13.75%_T\$275K_10y_ C±19_HE=1.05_PS=2	PI	\$275K/10yr	\$27,500/yr	5.7%	\$1,355,539	1.286
13	PI_12.50%_T\$250K_10y_ C±19_HE=1.05_PS=2	PI	\$250K/10yr	\$25,000/yr	5.2%	\$1,252,798	1.270
14	PI_13.75%_T\$275K_10y_ C±19_HE=1.05_PS=2	PI	\$275K/10yr	\$27,500/yr	6.0%	\$1,359,158	1.270
15	PI_13.75%_T\$275K_10y_ C±19_HE=1.05_PS=2	PI	\$275K/10yr	\$27,500/yr	6.4%	\$1,431,850	1.284
16	PI_12.50%_T\$250K_15y_ C±19_HE=1.05_PS=2	PI	\$250K/15yr	\$16,666/yr	6.3%	\$1,499,738	1.202
17	PI_12.50%_T\$250K_15y_ C±19_HE=1.05_PS=2	PI	\$250K/15yr	\$16,666/yr	7.0%	\$1,572,672	1.205
18	PI_12.50%_T\$250K_20y_ C±19_HE=1.05_PS=2	PI	\$250K/20yr	\$12,500/yr	6.8%	\$1,572,006	1.163
19	PI_12.50%_T\$250K_15y_ C±19_HE=1.05_PS=2	PI	\$250K/15yr	\$16,666/yr	6.7%	\$1,502,864	1.186
20	PI_12.50%_T\$250K_25y_ C±19_HE=1.05_PS=2	PI	\$250K/25yr	\$10,000/yr	5.8%	\$1,287,243	1.180
21	PI_12.50%_T\$250K_25y_ C±19_HE=1.05_PS=2	PI	\$250K/25yr	\$10,000/yr	6.4%	\$1,419,087	1.177
22	PI_12.50%_T\$250K_25y_ C±19_HE=1.05_PS=2	PI	\$250K/25yr	\$10,000/yr	7.3%	\$1,630,276	1.158
23	PI_12.50%_T\$250K_25y_ C±19_HE=1.05_PS=2	PI	\$250K/25yr	\$10,000/yr	7.7%	\$1,635,779	1.141
24	PI_12.50%_T\$250K_20y_ C±19_HE=1.05_PS=2	PI	\$250K/20yr	\$12,500/yr	7.8%	\$1,644,243	1.142
25	PI_12.50%_T\$250K_20y_ C±19_HE=1.05_PS=2	PI	\$250K/20yr	\$12,500/yr	8.3%	\$1,646,136	1.122

Detailed Comparison

Metric	Baseline	RE Grade	LMI Grade	15yr (LMI)	20yr (LMI)	25yr (LMI)
Total Annuity	\$250,000	\$250,000	\$250,000	\$250,000	\$250,000	\$250,000
Payout	10yr @ \$25,000	10yr @ \$25,000	10yr @ \$25,000	15yr @ \$16,666	20yr @ \$12,500	25yr @ \$10,000
Mortgage Type	IO	PI	PI	PI	PI	PI
Collar	±20%	±20%	±20%	±20%	±20%	±20%
Holiday Entry	0.90	1.05	1.05	1.05	1.05	1.05
PS Config	25% q5	20% q3	25% q3	20% q3	20% q3	20% q5
FP Margin	0.25%	0.25%	0.25%	0.25%	0.25%	0.25%
PoC Year 30	17.9%	4.9%	5.3%	6.3%	6.8%	5.8%
PoC Year 20	28.3%	12.2%	12.9%	13.5%	13.8%	10.0%
PoC Year 15	36.1%	19.1%	19.8%	19.0%	16.1%	13.1%
Mean Surplus	\$1,381,981	\$1,440,007	\$1,078,273	\$1,442,180	\$1,501,204	\$2,723,878
P5 Surplus	\$-825,657	\$8,607	\$-30,359	\$-154,075	\$-172,708	\$-82,676
Sharpe	0.829	1.333	1.363	1.202	1.163	1.180
FP Revenue	\$823,670	\$1,428,338	\$1,505,910	\$1,499,738	\$1,572,006	\$1,287,243
Profit Share	\$672,058	\$1,287,647	\$1,372,711	\$1,350,660	\$1,413,231	\$1,097,490
FP Margin	\$151,612	\$140,691	\$133,199	\$149,078	\$158,775	\$189,753
Funder Surplus	\$746,484	\$735,582	\$555,304	\$740,874	\$769,568	\$1,377,267
Premium	\$44,472	\$12,485	\$12,956	\$15,855	\$15,199	\$12,284
Final Mortgage	\$1,600,000	\$0	\$0	\$0	\$0	\$0
Protection	82.1%	95.1%	94.7%	93.7%	93.2%	94.2%

Conclusions

- **P&I + holiday entry 1.05 are the dominant levers:** These two changes alone reduce PoC from 17.9% to ~5%. The collar stays at $\pm 20\%$ (baseline) and FP margin at 0.25% (baseline).
- **Only one configuration meets reinsurance grade:** PI, \$250K/10yr, $\pm 20\%$, HE=1.05, PS 20% q3, FM 0.25% — at PoC 4.9%. The reinsurance threshold is tight with a $\pm 20\%$ collar.
- **Payout duration trade-off (fixed total):** At \$250K total, 10yr payout (5.3% PoC) outperforms 25yr (5.8% PoC) because the shorter annuity period leaves more years for the P&I amortisation to reduce the mortgage. All durations remain within LMI threshold, so the choice is borrower preference: higher annual income (10yr) vs longer income stream (25yr).
- **3-year profit share boosts FP revenue significantly:** Switching from q5 to q3 adds ~\$200K–\$300K in cumulative FP revenue, at the cost of ~0.5pp PoC. The 5-year period is preferred for reinvestment compounding but generates less revenue.
- **IO is not competitive under these constraints:** With collar $\leq \pm 20\%$, no IO configuration meets the reinsurance threshold. IO requires a wider collar to offset the permanent mortgage balance.

Methodology

76 Phase 1 + 5760 Phase 2 = 5,836 scenarios $\times$ 10K paths; 32 validated at 50K paths. Seed: 42.